

LC 473: Matchsticks to Square

Targeted Companies: PhonePe, Amazon

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1. Problem Setup & Target

Given an array of matchsticks, e.g., `[1, 1, 2, 2, 2]`, we need to return true or false depending on whether they can form a perfect square.

Core Logic:

Sum = 8

Each side = $(8 / 4) = 2$

If $\text{sum} \% 4 \neq 0 \rightarrow \text{return false}$

2. Approach & Backtracking

Let current sides = (0, 0, 0, 0)

Suppose target = 5, pick the first stick = 3. We have 4 choices to place it:

3	0	0	0
SIDE 1	SIDE 2	SIDE 3	SIDE 4

OR

0	3	0	0
SIDE 1	SIDE 2	SIDE 3	SIDE 4

OR

0 SIDE 1	0 SIDE 2	3 SIDE 3	0 SIDE 4
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Next stick = 2 → again 4 choices.

Hence, backtracking is necessary as at some point no stick can be placed without exceeding the target length.

```
state = dfs(idx) → trying to place stick number idx
```

3. Steps & Optimization

- **S1:** matchstick[idx] try placing into side 1, side 2, ... side 4
- **S2:** when idx == matchstick.size(), every stick is placed successfully → return true

Optimization: Try placing bigger matchsticks first. Sorting them in descending order prunes invalid paths much faster.

4. C++ Implementation

```
class Solution {
public:

    int n;
    int target;

    bool dfs(vector<int>& matchsticks, int idx, vector<int>&
sides){

        if(idx == n) return true;

        //SEB principle
        for(int i = 0; i < 4; i++){ //explore

            if(sides[i] + matchsticks[idx] > target){
```

```

        continue;
    }

    sides[i] += matchsticks[idx]; //seen
    if(dfs(matchsticks, idx + 1, sides)){
        return true;
    }

    sides[i] -= matchsticks[idx]; //backtrack
}

return false;
}

bool makesquare(vector<int>& matchsticks) {
    //basic idea ye he ki
    //initially har ek side empty hoga
    //phir ek ek krke matchstick place kreng
    //har ek side try kreng matchstick place krne ke liye
    //agar kisi point pe saare khtm ho gye to true
    //agar kisi point pe kisi bhi side me matchstick nhi daal
    paa rhe to
    //backtrack and try different side now to put previous
    stick

    n = matchsticks.size();
    sort(matchsticks.begin(), matchsticks.end(), greater<int>());

    //pehle bade matchsticks adjust krke dekhenge

    int sum = accumulate(matchsticks.begin(),
matchsticks.end(), 0);
    if(sum % 4 != 0) return false;
    target = sum / 4;

    vector<int> sides(4, 0); //4 sides initialized to 0

    return dfs(matchsticks, 0, sides); //0 mean 0th index se
    start kreng uthana
}
};

```